

Determine the thermal-equilibrium electron and hole concentrations in silicon at $T=300$ K for doping concentrations. (a) Let $N_d=10^{16}$ /cm³ and $N_a=0$. (b) Let $N_d=5 \times 10^{15}$ /cm³ and $N_a=2 \times 10^{15}$ /cm³. Assume that $n_i=1.5 \times 10^{10}$ /cm³ in silicon at $T=300$ K	1
Calculate the thermal-equilibrium electron and hole concentrations in Germanium for a given doping concentration. Consider a germanium sample at $T=300$ K in which $N_d=2 \times 10^{14}$ /cm³ and $N_a=0$. Assume that $n_i=2.4 \times 10^{13}$ /cm³.	2
Calculate the thermal-equilibrium electron and hole concentrations in a compensated p-type semiconductor. Consider a silicon semiconductor at $T=300$ K in which $N_a=10^{16}$ /cm³ and $N_d=3 \times 10^{15}$ /cm³. Assume $n_i=1.5 \times 10^{10}$ /cm³.	3
Consider silicon at $T=300$ K. Calculate the thermal-equilibrium electron and hole concentrations for impurity concentrations of (a) $N_a=4 \times 10^{16}$ cm³, $N_d=8 \times 10^{15}$ /cm³ and (b) $N_a=N_d=3 \times 10^{15}$ cm³ Recall that $n_i=1.5 \times 10^{10}$ /cm³	4
A compensated silicon material at $T=300$ K has impurity doping concentrations of $N_a=2.8 \times 10^{17}$ /cm³ and $N_d=8 \times 10^{16}$ /cm³. The mobility of electron=500cm²/V-s and hole mobility=300cm²/V-determine the (a) conductivity, (b) resistivity and (c) drift current density if applied electric field is 10 V/cm. $n_i=1.5 \times 10^{10}$ /cm³	5